

§11.5 - Alternating Series

Example #1 (Alternating Harmonic Series)

The sum

$$\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k}$$

exists.

Theorem #1 (Alternating Series Test)

Suppose (b_n) satisfies

$$\lim_{n \rightarrow \infty} b_n = 0 \quad \text{and} \quad b_{n+1} \leq b_n \text{ for all } n$$

then

$$\sum (-1)^n b_n$$

converges.

Remark

We will see later that

$$\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} = \ln(2).$$

Example #2

Determine if

$$\sum_{k=2}^{\infty} (-1)^k \frac{\ln(k)}{k^2}$$

converges.

Theorem #2 (Error Estimate for the AST)

Assume

$$S = \sum_{k=1}^{\infty} (-1)^{k+1} b_k$$

converges by the Alternating Series Test.

Then

$$|S - S_n| \leq b_{n+1}$$

where S_n is the n th partial sum of S .

Example #3

Choose $n \in \mathbf{N}$ such that

$$|S - S_n| \leq 10^{-4}$$

where

$$S = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k^2}.$$

Example #4

Compare

$$\ln(2) = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} + \dots$$

to

$$B = 1 - \frac{1}{2} - \frac{1}{4} + \frac{1}{3} - \frac{1}{6} - \frac{1}{8} + \frac{1}{5} - \frac{1}{10} - \dots$$